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SUPERSEDING
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MILITARY SPECIFICATION

TELESCOPE, STRAIGHT: M90 SERIES, M90A2 AND M90A3

Reactivated after February 23, 2006 and may be used for new and existing designs and acquisitions.

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC) and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers straight telescopes of three types, designated the M90 series, M90A2 and M90A3. The differences are reticle pattern and reticle illumination.

1.2 Classification. The Telescope is primarily intended for either a recoilless rifle, or direct fire of the howitzer and there are the following types:

1.2.1 Telescope: M90C. This Telescope is intended for use on the 105MM Recoilless Rifle: M27A1 and does not use tritium.

1.2.2 Telescope: M90D. This Telescope is intended for use on the 75MM Recoilless Rifle: M20 and does not use tritium .

1.2.3 Telescope: M90F. This Telescope is intended for alignment of the Hawk missile launcher and utilizes a plain cross line reticle. It is also used in conjunction with Radar Chronograph: M36 for alignment purposes and does not use tritium.

1.2.4 Telescope: M90A2. This Telescope is intended for use as the direct fire sight for the 105MM Light Howitzer: M119A1 and uses tritium.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, ATTN: AMSRD-AAR-AIS-SS, Picatinny, New Jersey 07806-5000 or emailed to ardec-stdzn@pica.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

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1.2.5 Telescope: M90A3. This Telescope is intended for use as the direct fire sight for the 105MM Light Howitzer: M119A2.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all requirements of documents cited in sections 3 or 4 of this specification, whether or not documents are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-13830	-	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	-	Fire Control Materiel; General Specification Governing the Manufacture, and Inspection
MIL-I-45607	-	Inspection of Equipment, Acquisition, Maintenance and Disposition of

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Markings for Shipment and Storage
MIL-STD-1916	-	DOD Preferred Methods for Acceptance of Product
MIL-STD-2073-1	-	DOD Materiel Procedures for Development and Application of Packaging Requirements

(Copies of federal and military specifications, standards, and handbooks are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

7680673	-	Fixture, Pressure Testing
7694010	-	Test Fixture for Telescope, M90 Series
8205096	-	Telescopes, Straight: M90C, M90D, and M90F
12599180	-	Telescope, Straight: M90A2
12984673	-	Telescope, Straight: M90A3

(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000)

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO 9000 - Quality Management and Quality Assurance Vocabulary

(Application for copies should be addressed to ANSI New York City Office Operations, 25 West 43rd Street, 4 floor, New York, NY 10036 or go online to http://www.ansi.org)

DEPARTMENT OF DEFENSE, DEFENSE SUPPLY AGENCY

DSAM 4145.8 - Radioactive Commodities in the DOD Supply System

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.1 Deleted.

3.1.2 Regulations pertaining to radioactive devices. Marking, identification, handling and shipping of radioactive devices shall be in accordance with DSAM 4145.8 and MIL-STD-129 as applicable.

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3.1.3 External tritium contamination. Tritium shall not be present on the external surfaces of the M90A2 Telescope in excess of 1000 dpm/100 sq. cm.

3.2 Materials. Materials shall be in accordance with drawings, material specifications, and general specifications forming a part of this specification.

3.3 Fabrication. The M90C, M90D and M90F telescopes shall be manufactured in accordance with Drawing 8205096 and drawings pertaining thereto. The M90A2 telescope shall be manufactured in accordance with Drawing 12599180 and drawings pertaining thereto. The M90A3 telescope shall be manufactured in accordance with Drawing 12984673 and drawings pertaining thereto.

3.4 General specification. The contractor shall be responsible for the compliance with requirements of specifications MIL-PRF-13830 and MIL-F-13926 as they apply.

3.5 Environmental.

3.5.1 Storage temperature. The telescope shall show no evidence of physical failure when thermally stabilized at ambient temperatures of +160 °F (+71 °C) ± 5 °F (±3 °C) and -60 °F (-51°C) ± 5 °F (± 3 °C) for a period of 4 hours ± 15 minutes at each temperature.

3.5.2 Operating temperature. The telescope shall meet the applicable requirements of this specification while exposed and thermally stabilized at ambient temperatures of +145°F (+63°C) ± 5°F (±3°C) and -50 °F (-46°C) ± 5°F (± 3°C) for a period of 4 hours ± 15 minutes at each temperature. Upon return to standard ambient temperature, +60 °F to +90°F (+16°C to + 32°C), from each operating temperature, the telescope shall meet the requirements of this specification.

3.5.3 Vibration.

3.5.3.1 Vibration “A”. Where required, the telescope shall withstand a total of 270 minutes ± 5 minutes of sweep-cycle vibration. The vibration shall be applied for 90 minutes ± 2 minutes along each of the three mutually perpendicular major axes. A complete sweep-cycle shall consist of vibration from origin, 5 hz at 1 inch double amplitude to mid-point 5 g’s ± 0.5 g’s at 500 hz to origin, and shall have a duration of 15 minutes ± 1 minute. Double amplitude shall be constant at 1 inch between 5 hz and 10 hz, and varied with frequency to maintain a constant 5 g’s ± 0.5 g’s acceleration between 10 hz and 500 hz. Upon completion of vibration, the telescope shall exhibit no evidence of damage and shall meet the requirements of this specification.

3.5.3.2 Vibration “B”. The telescope shall be vibrated in a vertical plane for a total of 15 minutes +/- 1 minute of sweep cycle vibration. A complete sweep cycle shall consist of vibration from origin (5 hz at 1 inch double amplitude) to mid-point (5 g’s +/- 0.5 g’s at 500 hz) to origin and shall have a duration of 15 minutes +/- 1 minute. Double amplitude shall be constant at 1 inch between 5 hz and 10 hz and varied with frequency to maintain a constant 5 g’s +/- 0.5 g’s acceleration between 10 hz and 500 hz. Upon completion of

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vibration, the telescope shall exhibit no evidence of damage or physical failure and shall meet the requirements of this specification.

3.5.4 Rain test. The Telescope shall meet the requirements of Procedure I of the rain test of MIL-F-13926.

3.6 Cleanliness. Subsequent to vibration the telescope shall meet the cleanliness requirements of MIL-PRF-13830. Inspection for dirt and foreign particles (classed as dirt) shall be made from the eye end only. The maximum size of any particle appearing on the reticle surface shall not exceed the width of the reticle line. The maximum number of dirt particle appearing within a central field of view of 50 mils diameter shall be three (3). The maximum number of dirt particles appearing outside this field of view shall be three (3). The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than ¼ of the reticle line width shall be ignored regardless of distribution.

3.7 Sealing.

3.7.1 Telescope: M90C, M90D and M90F. The telescope shall show no evidence of leakage when subjected to an external pressure of 5 ± 0.10 pounds per square inch gauge pressure (psig).

3.7.2 Telescope: M90A2 and M90A3. The M90A2 and M90A3 telescope body shall be capable of sustaining an internal pressure of 5.0 ± 0.50 pounds per square inch gauge pressure (psig) for a period of 2 hours (minimum) without loss of pressure in excess of 0.5 psig.

3.8 Purging, telescope: M90A2 and M90A3. The M90A2 and M90A3 telescope body shall be purged with dry nitrogen supplied at a pressure of 8 ± 0.50 psig until the dew point of the emergent gas is no higher than -25 degrees F. Purging apertures shall be quickly and positively closed to minimize intermingling of external atmosphere upon disconnecting of purging equipment.

3.9 Optical.

3.9.1 Collimation.

3.9.1.1 Open sight. The line of sight of the open sight shall be parallel with the optical axis of the telescope within one degree.

3.9.1.2 Telescope: M90C. The line of sight thru the boresight point shall subtend an angle of 51.7 mils ± 0.5 mils with a line of sight established parallel to the bottom of the dovetail mounting surface.

3.9.1.3 Telescope: M90D. The line of sight thru the boresight point shall subtend an angle of 41.5 mils ± 0.5 mils with a line of sight established parallel to the bottom of the dovetail mounting surface.

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3.9.1.4 Telescope: M90F. The line of sight thru the intersection of the cross lines shall be parallel to the bottom of the dovetail mounting surface within 0.5 mils.

3.9.1.5 Telescope: M90A2 and M90A3. The line of sight thru the boresight point shall subtend an angle of 30.0 mils ± 0.5 mils with a line of sight established parallel to the bottom of the dovetail mounting surface.

3.9.2 Image tilt. The image of a vertical target line shall be vertical within 1 degree of arc.

3.9.3 Parallelism of reticle and image.

3.9.3.1 Telescopes: M90C and M90D. With the flat of the dovetail horizontal, the horizontal reticle line through the geometric axis shall not deviate from the image of a horizontal target line by more than the angle subtended by a 3.0 mil block at one extremity of the horizontal reticle line when the other extremity of the reticle line is in coincidence with the horizontal target line.

3.9.3.2 Telescope M90F. With the flat of the dovetail horizontal, the horizontal reticle line through the geometric axis shall not deviate from the image of the horizontal target line by more than the angle subtended by a 3.0 mil block at one extremity of the horizontal reticle line when the other extremity of the reticle line is in coincidence with the horizontal target line.

3.9.4 Reticle accuracy, telescopes: M90C, M90D, M90A2 AND M90A3. The total angular subtense of the reticle graduations between zero and maximum range shall be as follows:

Telescope	Range	Mils	Tolerance (Mils)
M90C	0-2700	111.6	+/-0.6
M90D	0-2700	120.3	+/-0.6
M90A2	-------	60.0	+/-0.6
M90A3		60.0	+/-0.6

3.9.5 Parallax. Parallax at the optical axis shall not exceed 0.2 mil when viewing a target at a distance of between 200 yards and infinity.

3.9.6 Resolution. Resolving power on the optical axis shall be 20 seconds of arc or less.

3.9.7 Eyepiece focus. The eyepiece focus shall be fixed between -0.50 and -1.0 diopter.

3.9.8 Illumination. The telescope shall meet the “Illuminated reticles” requirement of MIL-PRF-13830. The reticle markings shall be clearly distinguishable when observed

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in ambient light conditions, ranging from dusk to darkness.

3.10 Eyeshield. The rubber eyeshield shall show no evidence of deterioration, cuts, or tears and shall fit snugly on the telescope body.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in ISO 9000.

4.2 First article (initial production) approval. The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval of test shall consist of three (3) telescopes plus three (3) each of all items covered by Supplementary Quality Assurance Provisions (SQAP). The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing. When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all requirements of the contract and submit a record of this inspection, with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing. When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for apporval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection provisions.

4.3.1 Submission of product. Unless otherwise specified by the contracting office, inspection lot size, lot formation and presentation of lots shall be in accordance with “Submission of Product” provisions of MIL-STD-1916.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies. All components and subassemblies shall be inspected in accordance with the inspection provisions contained in SQAP listed in the Technical Data Package (TDP). In the absence of SQAP's the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final acceptance inspection. Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristics) basis in accordance with MIL-STD-1916 and the sampling plans specified in Table I herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-STD-2073-1 and Section 5 herein. The tabulated classification of defects in Table I and II shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I – Classification of defects.

NO	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
MAJOR: 100% INSPECTION			
101	Sealing (M90C, M90D, M90F)	3.7.1	4.6.5.1
102	Sealing (M90A2, M90A3)	3.7.2	4.6.5.2
103	Purging (M90A2, M90A3)	3.8	4.6.6
MAJOR: VERIFICATION LEVEL IV			
104	External Tritium Contamination	3.1.3	4.6.1
105	Vibration	3.5.3	4.6.2.2
106	Cleanliness	3.6	4.6.4
107	Open Sight	3.9.1.1	4.6.6
108	Collimation (M90C)	3.9.1.2	4.6.6
109	Collimation (M90D)	3.9.1.3	4.6.6
110	Collimation (M90F)	3.9.1.4	4.6.6
111	Collimation (M90A2, M90A3)	3.9.1.5	4.6.6
112	Image Tilt	3.9.2	4.6.7
113	Parallelism of Reticle & Image (M90C & M90D)	3.9.3.1	4.6.8
114	Parallelism of Reticle & Image (M90F)	3.9.3.2	4.6.8
115	Reticle Accuracy (M90C, M90D, M90A2 & M90A3)	3.9.4	4.6.9
116	Parallax	3.9.5	4.6.10
117	Resolution	3.9.6	4.6.11
118	Eyepiece focus	3.9.7	4.6.12
119	Illumination	3.9.8	4.6.13
120	Eyeshield	3.10	4.7

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MINOR: NONE

NOTE: The tests of Table I shall be conducted at a temperature between +60 degrees and +90 degrees F

4.3.3 Acceptance and rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejections" as specified in MIL-STD-1916.

4.4 Special sampling.

4.4.1 General. Three telescopes shall be selected at random as a special sampling from each 100 produced or one months production, whichever occurs first. The sample shall meet the requirements of Table II.

TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301	Fabrication	3.3	Applicable drawings – visual
302	General Specification	3.4	MIL-F-13926 & MIL-PRF-13830
303	Storage Temperature	3.5.1	4.6.2.1
304	Operating Temperature	3.5.2	4.6.2.1
305	Rain	3.5.4	4.6.2.3

4.4.2 Failure of sample. Should any one of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Test equipments.

4.5.1 Inspection equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.2 Government furnished inspection equipment. Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.3 Contractor furnished inspection equipment.

4.5.3.1 Government design. Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the list of inspection documents when provided with the TDP.

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4.5.3.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance limit used in the design of the test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3.3 Inspection equipment design requirements.

4.5.3.3.1 Sealing, telescopes: M90C, M90D, & M90F. Special pressure testing equipment, as shown on Drawing 7680673, shall be capable of being rendered pressure-tight and of holding the required pressure for the specific period of time with a test specimen inserted in the chamber.

4.5.3.3.2 Performance. Special test equipment utilized to test the telescope shall conform to drawing 7694010, Test Fixture for Telescope, M90 Series. The test equipment shall permit inserting and securing the dovetail of the telescope in a horizontal plane within 10 seconds of arc. The fixture shall also consist of a target collimator containing a calibrated reticle to check collimation, parallax and accuracy readings.

4.6 Test methods and procedures.

4.6.1 External tritium contamination. Contamination test shall be performed by wiping the M90A2 telescope with filter paper moistened in distilled water. The wiping shall be performed with moderate finger pressure. The damp filter paper shall be placed in a container with the proper portion of scintillation liquid. The read out shall be within 10% of the known standard value when counted to a total of 2000 disintegrations. The actual test for determining contamination is performed by placing the container into the scintillation system. The measured contamination shall be within the limits specified in 3.1.3.

4.6.2 Environmental.

4.6.2.1 Storage and operating temperatures. Test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.5.1. The telescope shall be exposed to and stabilized at the temperature extremes for a minimum of 6 hours. The temperatures shall be cycled to avoid subjecting the telescope to thermal shock. Subsequent to the temperature exposures, the telescope shall be stabilized in an ambient temperature between +60 and 90 F and subjected to the requirements of 3.5.2.

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4.6.2.2 Vibration. Standard vibrating equipment capable of providing the amplitude and frequency specified in 3.5.3 shall be utilized for the vibration test. Unless otherwise specified, the test equipment shall be in accordance with the "test facilities" requirements of MIL-F-13926. The telescope shall be mounted horizontally by means of a government approved adapter that is secured to the vibration fixture. This adapter shall contain the necessary dovetail slot capable of mounting and securing the dovetail of the telescope during the vibration test. Subsequent to the test the telescope shall be visually examined for damage. The telescope shall then meet the examinations and tests of this specification.

4.6.2.3 Rain. The rain test chamber shall be in accordance with the "Test Facilities" section of MIL-F-13926. The test shall be performed according to Procedure I of the rain test in MIL-F-13926. There shall be no degradation of the telescope and it shall meet the examinations and tests of this specification.

4.6.3 Cleanliness. The optical surfaces of the telescope shall be inspected for cleanliness in accordance with MIL-PRF-13830. The telescope shall conform to the requirement of 3.6.

4.6.4 Sealing.

4.6.4.1 Sealing, telescopes: M90C, M90D & M90F. The external pressure testing of the telescope shall be performed with test equipment conforming to 4.5.3.3.1. The pressure shall be applied to the telescope by means of the adapter specified in the test equipment. Place the telescope in the adapter and pressurize with dry air.

Retain the applied pressure of 5 ± 0.10 pounds per square inch gage (psig) on the telescope for a period of 5 minutes ± 15 seconds. At the end of the time period there shall be no evidence of leakage as detected on the dial indicators of the testing equipment, in order to conform to the requirements of 3.7.1.

4.6.4.2 Sealing, telescopes: M90A2 & M90A3. The gage used in this test shall be capable of assuring conformance to the requirements of 3.7.2. The telescope shall be pressurized with dry nitrogen to a pressure of 5.0 +/- 0.5 pounds per square inch gage. The pressure shall be noted and subsequent to the 2 hour period checked for compliance with 3.7.2.

4.6.5 Purging telescopes: M90A2 & M90A3. Connect the test equipment to the purging valve of the telescope. Adjust the dry nitrogen to the pressure specified in 3.8. The pressure is to remain constant until the dewpoint of the emergent gas is no higher than -25 degrees F.

4.6.6 Collimation. The collimation test shall be performed using the test equipment described in 4.5.3.3.2 or by using a government approved wall target. The wall target shall be placed at a distance and have target lines that shall represent the applicable requirements of 3.9.1. The test collimator or wall target must locate the line of sight parallel to the bottom of the dovetail to within 10 seconds. Once the bottom of the dovetail surface is

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located, sight through the telescope eyepiece to determine if the boresight point is in conformance with 3.9.1.

4.6.7 Image tilt. This test shall be performed with the aid of an auxiliary telescope containing a vertical and horizontal reticle line, and the test equipment in 4.5.3.3.2. The telescope shall be at least 3 power (3X). Plumb the vertical reticle line of the telescope to a plumb line and position the eyepiece of the telescope under test. View through the telescope and establish one extremity of the telescope vertical reticle line coincident to the respective target image line. The image of the target shall be parallel to the telescope's reticle line within the tolerance specified in 3.9.2 as measured on the target.

4.6.8 Parallelism of reticle and image. This test is made with one extremity of the horizontal reticle line of the auxiliary telescope in coincidence with the horizontal target line image. The reticle of the telescope under test shall be parallel to the target line within the tolerances specified in 3.9.3.1 and 3.9.3.2 as measure on the target.

4.6.9 Reticle accuracy, telescopes M90C, M90D, M90A2 & M90A3. Utilizing the equipment in 4.5.3.3.2 the horizontal cross line of the boresight line shall be coincident to a calibrated mark on the target reticle. A government approved wall target can also be used in place of the collimator target. The maximum range lines on the reticle shall meet the requirements and shall be within the tolerance specified in 3.9.4 as measured on the target image.

4.6.10 Parallax. This test shall be performed by observing through the telescope eyepiece. Parallax is recognized as displacement caused between the telescope reticle and the image of the target when the observer's head is moved from side to side or up and down. Such movement of the observer's head shall not cause displacement of the two images to exceed the tolerance specified in 3.9.5 at the distance indicated.

4.6.11 Resolution. A resolution target and an auxiliary telescope of at least 3 power (3X) shall be used for this test. The resolution target shall be either a wall target or reticle in a collimator. The target must represent the angular subtense for seconds of arc specified in 3.9.6. The target used shall either be as specified in MIL-PRF-13830 or the Air Force Resolution chart which forms a part of NBS circulars. It shall have the detectable line structure and proper line count in each meridian. The telescope shall resolve the target within the requirements of 3.9.6.

4.6.12 Eyepiece focus. This test shall be performed utilizing a dioptomer with at least a minus 1.0 diopter to a plus 1.0 diopter setting. The crossline reticle of the dioptometer shall be adjusted to the individual eye. Placing the dioptometer at the eyepiece of the telescope under test, observe the sharpness of the telescope reticle in comparision to the reticle in the dioptometer. Adjust the dioptometer until both reticle lines appear to be equal, then read the diopter scale of the dioptometer. The index shall be within the tolerance specified in 3.9.7.

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4.6.13 Illumination. This is a visual check performed in a darkened area. The graduations shall appear to be evenly illuminated without flaring at the edge of the graduations or bright spots over the area as required by 3.9.8.

4.7 Eyeshield. The eyeshield shall be given a visual and tactile examination. The material shall show no physical damage or deterioration and it shall fit snug to the eyepiece end of the telescope as required by 3.10.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency or within the military service's system commands. Packaging data retrieval is available from the managing Military Department’s or Defense Agency's automated packaging files, CD ROM products, or by contacting the responsible packaging activity.

6. NOTES

6.1 Intended use.

6.1.1 Telescope: M90C. This Telescope is intended for use on the 105MM Recoilless Rifle: M27A1.

6.1.2 Telescope: M90D. This Telescope is intended for use on the 75MM Recoilless Rifle: M20.

6.1.3 Telescope: M90F. This Telescope is intended for alignment of the Hawk missile launcher and utilizes a plain cross line reticle. It is also used in conjunction with Radar Chronograph: M36 for alignment purposes.

6.1.4 Telescope: M90A2. This Telescope is intended for use as the direct fire sight for the 105MM Light Howitzer: M119A1.

6.1.5 Telescope: M90A3. This Telescope is intended for use as the direct fire sight for the 105MM Light Howitzer: M119A2.

6.2 Ordering data. Procurement data should specify:

a. Title, date and number of this specification.
b. Serial numbers to be used for identification marking.
c. Applicable packaging data sheet number.
d. Selection of an applicable level of preservation and packaging.

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e. Requirements for first article testing.
f. Contract data requirements for submission of inspection equipment designs to Data Item Description DI-R-1714.

6.3 Submission of designs for approval. Contractor designs for final acceptance inspection must be approved by the Government prior to fabrication or procuring the equipment. The contractor is referred to MIL-HDBK-204 for guidance. Submission of design concept on inspection equipment is permissible for tentative approval. The completion date for design review will be based on the date of the final submission of designs and the required delivery schedule as stipulated in the contract. Submit designs as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAF-I (D), Picatinny Arsenal, NJ 07806-5000. When the contractor submits inspection equipment designs to the government for approval, he should give the following information in his letter of transmittal:

a. The contract number.
b. The contract item (name, model number, etc).
c. The designs remaining to be submitted and the expected date of submittal.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Subject term (key word) listing.

Telescope
Optical
Fire control
Radioactive
Tritium

6.6 Hazardous material.

6.6.1 Tritium. The M90A2 Telescope, Straight contains tritium, a radioactive material. Prospective bidders should be cognizant of the need for a Nuclear Regulatory Commission License governing the procurement, processing, handling and shipping of radioactive material.

6.6.2 Lithium Batteries. The M90A3 utilizes Lithium batteries. Prospective Bidders must be cognizant of the need to handle and/or dispose of these batteries in accordance with all Local, State and Federal Laws.

6.7 Packaging data sheet drawings. The following packaging data sheet drawings should be obtain from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIL-P B455, Picatinny, NJ 07806-5000 .

SPI AM7648432 - Special Packaging Instruction for Telescope Straight:

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		M90
SPI AM8205096	-	Special Packaging Instruction for Telescope Straight: M90
SPI AM12599180	-	Special Packaging Instruction for Telescope Straight: M90A2
SPI AM12984673	-	Special Packaging Instruction for Telescope Straight: M90A3

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Army-AR

Preparing activity:
Army-AR
(Project 1240-2006-003)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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